HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and maintained energy modelling studies focusing on **energy efficiency measures** and **heating system decarbonisation** options, contributing to national policy insights.
- Built and enhanced analytical tools using **Python** for energy system analysis, integrating data to improve model inputs and outputs.
- Prepared, integrated, and analysed diverse datasets to refine energy modelling parameters and forecast outcomes, directly impacting the accuracy of research findings.
- Authored comprehensive technical reports and data visualisations, translating complex analytical findings on energy systems for both technical and non-technical stakeholders.
- Researched emerging trends in energy systems and **decarbonisation**, identifying opportunities to enhance existing modelling methodologies and analytical tools.
- Supported the development of energy modelling frameworks and analytical processes, ensuring consistency and scalability for future research initiatives.
- Managed and maintained relationships with **internal and external stakeholders**, including presenting research findings and representing the organisation at industry meetings.

SKILLS

Programming Languages – Python, SQL, R

Data Analysis & Modelling – Energy Modelling, Statistical Modelling, Data Analysis, Data Visualisation, Data Integration, Exploratory Data Analysis

Tools & Technologies – Power BI, Tableau, pandas, NumPy, scikit-learn, Jupyter Notebooks, Git

Databases – PostgreSQL, MySQL, Oracle

Cloud & Infrastructure – AWS (EC2, S3, Lambda, CloudWatch), Docker, Linux

Methodologies – Agile, Scrum

PROJECTS

Energy System Efficiency Analysis Tool (MSc Project)

- Developed a **Python**-based analytical tool to model and assess the impact of various **energy efficiency measures** on building heating systems.
- Integrated diverse datasets including building specifications and historical energy consumption using **pandas** and **SQL** to create comprehensive input parameters.
- Generated detailed **data visualisations** and reports using **Power BI** to present findings on potential carbon reductions and cost savings to academic supervisors.

Decarbonisation Pathway Modeler (HCL Technologies)

- Contributed to the development of a modelling framework within an enterprise environment, focusing on analysing **heating system decarbonisation** options.
- Automated data ingestion and transformation processes using **Python** scripting and **SQL** queries against enterprise databases (PostgreSQL, Oracle).
- Supported the analysis and reporting of findings related to Ireland's transition to a low-carbon future, collaborating with operations and business analyst teams.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)